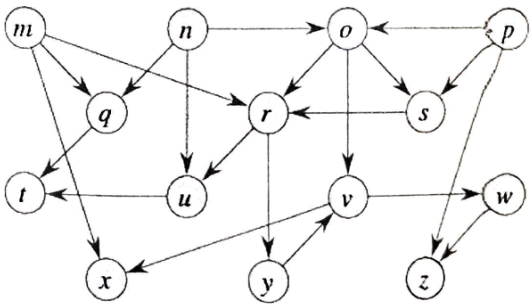
National Institute of Technology, Kurukshetra-136119

Mid Term- II, Odd Semester 2025-2026

Subject: Design and Analysis of Algorithm (CSPC 201/DSPC 203) **Time:** 50 Minutes

Maximum Marks: 15 **Branch:** B.Tech. M&C, AI-Data Science (III Semester)

1	Define the Minimum Spanning Tree (MST) problem with a suitable example computed using Kruskal's algorithm. Also, write Kruskal's algorithm for finding the MST and determine its time complexity.	6
2	Perform DFS on the following graph, starting from vertex 'm'. If not all vertices are visited through DFS from 'm', continue performing DFS from the remaining unvisited vertices in the order 'n', 'o', 'p', and so on, until all vertices in the graph have been visited. If you have two or more than two choices to move during traversal, choose vertex with smaller character among all choices (e.g. to move from vertex 'm', first traverse through 'q', then 'r', and finally 'x', if required). Clearly mention the discovery time and finish time of each vertex, and show the depth first forest. Also categories the edges of the graph in the tree edge, back edge, forward edge, and cross edge. 	6
3	What is optimal Huffman code for character "a, b, c, d, e, f, g, and h" with following set of frequencies, **a:21, b:1, c:5, d:3, e:2, f:13, g:8, h:1**	3